XIYUN HU

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SUMMARY

Innovative Ph.D. candidate in Mechanical Engineering with over 4 years of hands-on experience in machine learning and computer vision, specializing in augmented reality applications. Proficient in **Python**, **PyTorch**, and **OpenCV**. I've successfully led projects like GesPrompt, creating an **LLM**-powered **XR** copilot that enhances real-time interaction through gesture and voice recognition. My work has not only improved user experience but also contributed to the development of cutting-edge tools for interactive design and training in mixed reality environments.

EDUCATION

Purdue University — **Jan. 2021 - Aug. 2026 (Expected)**
Ph.D., Mechanical Engineering — West Lafayette, IN

Purdue University — **Aug. 2019 - Dec. 2020**
M.S., Mechanical Engineering — West Lafayette, IN

University of Pittsburgh — **Aug. 2017 - Apr. 2019**
B.S., Mechanical Engineering — West Lafayette, IN

Sichuan University — **Aug. 2015 - Apr. 2019**
B.Eng., Mechanical Engineering — Sichuan, China

PROFESSIONAL EXPERIENCE

Research Intern — **Sep. 2025 - Apr. 2026**
Fujitsu Research of America — Pittsburgh, PA (remote)

- Conducted a systematic survey of state-of-the-art human motion synthesis and video generation literature to identify research gaps and define a novel experimental direction.
- Developed a **3D human motion generation** pipeline that integrates **image generation**, **video generation**, and **human mesh recovery** to reconstruct temporally consistent 3D motion from text and 2D inputs.
- Designed and prototyped a user interface enabling interactive control of the motion generation pipeline and conducted evaluations of usability and motion quality.
- The project is submitted to UIST 2026.

SELECTED PROJECTS

ARify: Leveraging Narrated Instructional Videos to Create Augmented Reality Tutorials for Procedural Tasks.

- Designed and developed a semi-automatic AR authoring system that converts narrated instructional videos into spatially anchored AR tutorials, lowering the barrier for non-programmers to create procedural training content.
- Integrated perception modules for **object detection, segmentation, pose estimation, and hand tracking** with a **VLM** pipeline that parses instructional videos into hierarchical task → step → action structures and automatically maps instructional tactics to AR representations.
- Implemented a desktop AR authoring workflow for refining tutorial structures and spatial placement of AR guidance elements.

GesPrompt: Leveraging Co-Speech Gestures to Augment LLM-Based Interaction in Mixed Reality

- Designed and developed a multimodal processing workflow for XR interactions with LLM, allowing for more expressive user input through the integration of speech and gesture recognition.
- Enabled the effective joint processing of gesture and speech inputs by prompt-tuning on an **LLM (GPT4) agent**, improving interaction accuracy in XR applications.
- Achieved improved clarity in user commands by designing a **co-speech gesture framework and recognition** system that extracts and interprets spatial parameters from gestures, clarifying speech ambiguities.

agentAR: Creating Augmented Reality Applications with Tool-Augmented LLM-based Autonomous Agents

- Developed a **tool-augmented LLM agent** that streamlines the AR application development process, enabling users to create AR applications using natural language.
- Established a comprehensive **ML tool library** for agent-based AR application creation workflow by deriving a common structure and framework through reviewing state-of-the-art AR research.

- Confirmed the robustness of agentAR through comprehensive testing of AR applications by successfully recreating 6 existing AR applications and creating 6 new AR applications, each incorporating different **ML algorithm** combinations.

Ubi Edge: Authoring Edge-Based Opportunistic Tangible User Interfaces in Augmented Reality
- Designed and developed an AR system to detect **3D edges** in the physical environment, enabling users to author applications leveraging those edges.
- Created a 3D edge point cloud for objects by integrating through the integration of an **RGBD-based edge detection** algorithm.
- Increased the efficiency of point cloud segmentation by utilizing the **YOLO** object detection model to segment the point cloud of the object from the environment.
- Optimized edge point matching process by utilizing **BundleTrack** for **6D pose tracking** and applying the **edge-based ICP** method.

ARnnotate: An Augmented Reality Interface for Collecting Custom Datasets of 3D Hand-Object Interaction Pose Estimation
- Created an innovative AR-based workflow that enabled continuous and pervasive collection of custom **hand-object pose estimation datasets**, enhancing the efficiency of dataset collection for machine learning applications.
- Verified dataset quality by implementing **CenterPose** object pose estimation network and **OpenPose** hand detection networks using **TensorFlow** on a **Linux** environment.
- Increased dataset quality and optimized user experience through the use of an AR interface that synergizes front-end visual assistance with back-end computational processes using **Unity3D** on **Oculus Quest 2**.

Scalar: Authoring Semantically Adaptive Augmented Reality Experiences in Virtual Reality
- Designed an innovative AR/VR workflow that accurately defined and validated **semantically adaptive AR experiences** within synthetically generated VR environments for the advancement of immersive technology solutions.
- Achieved accurate **3D object detection** on HoloLens 2 by integrating a **3D semantic understanding** network alongside a **YOLO** object detection network.
- Implemented an **SVM-based** algorithm that effectively aligns with AR designer demonstrations, serving as a semantic adaptation model for seamless experience deployment.

Gesturar: An Authoring System for Creating Freehand Interactive Augmented Reality Applications
- Developed a novel workflow for customizing freehand interactive AR experiences to enhance user engagement through in-situ gesture demonstration and visual programming.
- Delivered a robust gesture detection system by implementing a **CNN for gesture detection** and a **Siamese network** for gesture comparison with PyTorch and Unity Barracuda.
- Created a user-centric interaction framework that facilitated mapping of gestural inputs to virtual content behaviors across 4 interaction modes.

SKILLS

Programming Skills: **Python**, **C#**, Git, Linux, Arduino
ML/DL Platforms & Libraries: **Pytorch**, TensorFlow, OpenCV, Keras, Scikit-learn, Transformers
AR/VR & Graphics Tools: **Unity3D**, Blender, MeshLab, PTC Creo, **Oculus**, **HoloLens**

ACADEMIC SERVICE

Reviewer: CHI (2023-2025), UIST (2025), CSCW (2024), IEEE VR (2025), DIS (2025)

HONORS AND AWARDS

Paper Honorable Mention: CHI 2026, CHI 2024, UIST 2021
Special Recognition for Outstanding Review: CHI 2025, DIS 2025, UIST 2025

PUBLICATIONS

Conference Papers

[c.15] ARify: Leveraging Narrated Instructional Videos to Create Augmented Reality Tutorials for Procedural Tasks.
Hu, X.*, Zhu, C*., Hsia, S.*, Ma, D., Jain, R., & Ramani, K. (2026, April).
In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3772318.3790715
[c.14] JustShape: Exploring Co-Speech Gestures for Multimodal LLM-Powered 3D Parametric Modeling.

Honorable Mention
Duan, R*, Chen, Y.*, Hu, Y., Liu, Z., Zhu, C., **Hu, X.**, Ma, D., Wang, X., & Ramani, K. (2026, April).
In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3772318.3790641

[c.13] AgentCoach: LLM-Based Adaptive Coaching Feedback for Motor Skill Learning.
Ma, D.*, Yu, J.*, Wang, X, **Hu, X.**, He, L., Jeong, S., & Ramani, K. (2026, April).
In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3772318.3791652

[c.12] AmIWrite: Exploring Scalable One-on-One Handwriting-Based Tutoring for Mathematical Problem-Solving with an LLM-Powered AI Tutor.
Liu, Z.*, Chen, Y.*, Ji, H., Duan, R., Zhu, Z, **Hu, X.**, Peppler, K., & Ramani, K. (2026, April).
In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3772318.3790935

[c.11] agentAR: Creating Augmented Reality Applications with Tool-Augmented LLM-based Autonomous Agents.
Zhu, C*., Hsia, S.*, **Hu, X.***, Liu, Z., Shi, J., & Ramani, K. (2025, September).
In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology.
DOI: https://doi.org/10.1145/3746059.3747676

[c.10] GesPrompt: Leveraging Co-Speech Gestures to Augment LLM-Based Interaction in Virtual Reality.
Hu, X.*, Ma, D.*, He, F., Zhu, Z., Hsia, S., Zhu, C., Liu, Z., & Ramani, K. (2025, July).
In Proceedings of the 2025 ACM Designing Interactive Systems Conference.
DOI: https://doi.org/10.1145/3715336.3735769

[c.9] AdapTUI: Adaptation of Geometric-Feature-Based Tangible User Interfaces in Augmented Reality.
He, F.*, **Hu, X.***, Qian, X., Zhu, Z., & Ramani, K. (2024, October).
In Proceedings of the 2024 ACM Interactive Surfaces and Spaces Conference.
DOI: https://doi.org/10.1145/3698127

[c.8] avaTTAR: Table Tennis Stroke Training with Embodied and Detached Visualization in Augmented Reality.
Ma, D.*, **Hu, X.***, Shi, J., Patel, M., Jain, R., Liu, Z., Zhu, Z., & Ramani, K. (2024, October).
In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology.
DOI: https://doi.org/10.1145/3654777.3676400

[c.7] Parametric: Empowering In-Situ Parametric Modeling in Augment Reality for Personal Fabrication.
Duan, R.*, **Hu, X.***, Liu, M., Shi, J., & Ramani, K. (2024, August).
In Proceedings of the ASME 2024 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDECT-CIE).
DOI: https://doi.org/10.1115/DETC2024-146420

[c.6] ClassMeta: Designing Interactive Virtual Classmate to Promote VR Classroom Participation.
Honorable Mention
Liu, Z., Zhu, Z., Zhu, L., Jiang, E., **Hu, X.**, Peppler, K. A., & Ramani, K. (2024, May).
In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3613904.3642947

[c.5] Ubi Edge: Authoring Edge-Based Opportunistic Tangible User Interfaces in Augmented Reality.
He, F.*, **Hu, X.***, Shi, J., Qian, X., Wang, T., & Ramani, K. (2023, April).
In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3544548.3580704

[c.4] ARnnotate: An Augmented Reality Interface for Collecting Custom Dataset of 3D Hand-Object Interaction Pose Estimation.
Qian, X., He, F., **Hu, X.**, Wang, T., & Ramani, K. (2022, October).
In Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology.
DOI: https://doi.org/10.1145/3526113.3545663

[c.3] Scalar: Authoring semantically adaptive augmented reality experiences in virtual reality.
Qian, X., He, F., **Hu, X.**, Wang, T., Ipsita, A., & Ramani, K. (2022, April).
In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3491102.3517665

[c.2] Gesturar: An authoring system for creating freehand interactive augmented reality applications.
Honorable Mention
Wang, T., Qian, X., He, F., **Hu, X.**, Cao, Y., & Ramani, K. (2021, October).
In The 34th Annual ACM Symposium on User Interface Software and Technology.
DOI: https://doi.org/10.1145/3472749.3474769

[c.1] CAPturAR: An augmented reality tool for authoring human-involved context-aware applications.
Wang, T., Qian, X., He, F., **Hu, X.**, Huo, K., Cao, Y., & Ramani, K. (2020, October).
In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology

DOI: https://doi.org/10.1145/3379337.3415815

Extended Abstracts

[e.1] Towards Design Guidelines for Natural-Language-Based Human-Multi-Robot Collaboration in Domestic Environments.
Wang, X*, Hsia, S.*, Liu, Z.*, Zhu, C., Zhu, Z., **Hu, X.**, Ostrowski, A., & Ramani, K. (2026, April).
In Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems.
DOI: https://doi.org/10.1145/3772363.3798976

Patents

[p.1] Karthik Ramani, Fengming He, Xun Qian, Jingyu Shi, Xiyun Hu. 2024. Authoring edge-based opportunistic tangible user interfaces in augmented reality. U.S. Patent Application No. 18/584,258.